

BPMN Process Modeling Assignment

Scenario 1: Employee Leave Approval

In this process, an employee wants to apply for leave through the company's HR system. The process starts when the employee submits the leave request. The HR system then checks the employee's available leave balance.

After checking the balance, an exclusive gateway is used to decide what happens next. If the employee has enough leave balance, the request is sent to the manager for approval. If there is not enough leave balance, the system sends an insufficient-balance notification to the employee and the process ends.

If the request is sent to the manager, the manager can either approve or reject it. If the manager approves the request, the system updates the employee's leave balance and sends an approval notification. If the manager rejects the request, a rejection notification is sent to the employee. In both cases, the process ends after the notification is sent.

BPMN Flow:

Start Event



Submit Leave Request



Check Leave Balance



[Leave Balance Sufficient?]

— Yes → Send Request to Manager



[Manager Approves?]

— Approved → Update Leave Balance



Approval Notification



End

— Rejected → Rejection Notification



End

— No → Insufficient-Balance Notification



End

Scenario 2: Online Purchase Order Processing

In this process, the customer starts by placing an order online. The system first checks whether the product is available. An exclusive gateway is used to decide the next step based on the product availability.

If the product is not available, the customer receives an out-of-stock notification and the process ends. If the product is available, the system continues with payment processing.

After the payment is processed, another exclusive gateway checks whether the payment was successful. If the payment fails, the customer receives a payment failure notification and the process ends. If the payment is successful, the system confirms the order and prepares the product for shipment. The product is then shipped and the customer receives a shipping confirmation. After this, the process ends.

BPMN Flow:

Start Event



Place Order



Check Product Availability



[Product Available?]

— No → Out-of-Stock Notification



— Yes → Process Payment



[Payment Successful?]

— No → Payment Failure Notification



— Yes → Confirm Order



Prepare Product



Ship Order



Shipping Confirmation



End

Scenario 3: IT Service Request

In this process, an employee reports an IT problem and submits an IT support request. The IT help desk first registers the request and then checks the severity of the problem.

An exclusive gateway is used to decide which technician should handle the request. If the problem is low severity, it is assigned to a support technician. If the problem is high severity, it is assigned to a senior technician. The technician then investigates the problem.

After investigating the problem, another exclusive gateway is used to check whether the problem can be solved internally. If it can be solved, the technician fixes the problem. If it cannot be solved internally, the problem is escalated to an external service provider.

After the issue is resolved, the help desk updates the request status. The employee then receives a resolution notification and the process ends.

BPMN Flow:

Start Event



Submit IT Support Request



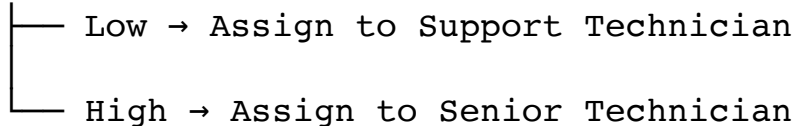
Register Request



Check Problem Severity



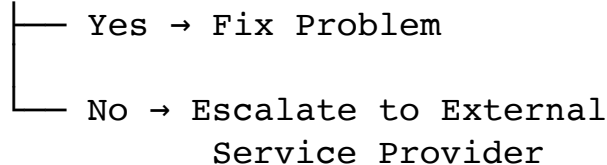
[Problem Severity?]



Investigate Problem



[Can Problem Be Resolved Internally?]



Update Request Status



Resolution Notification



End

